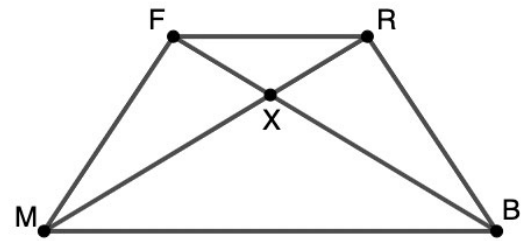


Geometry
Unit 8
Lesson 5 Practice

Name **Answer Key** _____
Date _____
Period _____

Given: $FRBM$ is an isosceles trapezoid, $\angle MFX \cong \angle BRX$.

Prove: $\triangle MBX$ is an isosceles triangle.



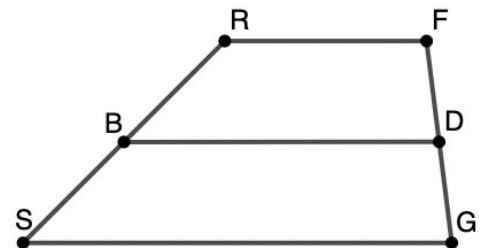
Statement	Reason
$FRBM$ is an isosceles trapezoid	Given
$\angle FMB \cong \angle RBM$	1. Base angles of an isosceles trapezoid are congruent
$\angle MFX \cong \angle BRX$	Given
2. $\overline{MB} \cong \overline{MB}$	Reflexive Property of Congruence
3. $\triangle FMB \cong \triangle RBM$	AAS Congruence
4. $\angle FBM \cong \angle RMB$	Corresponding Parts of Congruent Triangles are Congruent
$\triangle MBX$ is an isosceles triangle	5. Base angles of an isosceles triangle are congruent

Trapezoid $SRFG$ is shown. $\overline{BD}$ is a midsegment of $SRFG$.
 $FD = 4x + 34$, $DG = 8x - 14$, $RB = 8y + 41$, $SB = 13y - 4$.

6. Find the values of x and y .

$$x = 12$$

$$y = 9$$

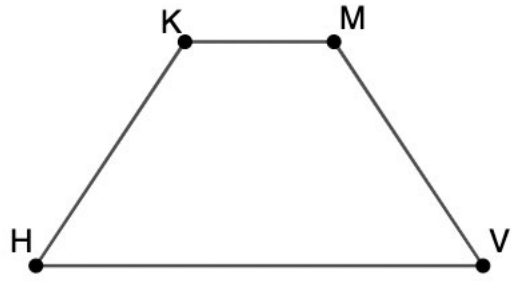


7. Find the lengths of $\overline{RS}$ and $\overline{DG}$.

$$RS = 226$$

$$DG = 82$$

Isosceles trapezoid $KMVH$ is shown where $m\angle KHV = (16z - 6)^\circ$, $m\angle HKM = (21z + 38)^\circ$, $m\angle KMV = (16t - 26)^\circ$.



8. What is the value of z ?

$$z = 4$$

9. What is the value of t ?

$$t = 6$$

10. What are $m\angle VHK$, $m\angle HKM$, $m\angle KMV$, and $m\angle MVH$?

$$m\angle VHK = 58^\circ$$

$$m\angle MVH = 58^\circ$$

$$m\angle HKM = 122^\circ$$

$$m\angle KMV = 122^\circ$$